
Complete HTML Notes – Deep Definitions

 For Students | Interviews | Projects | Beginner to Pro

◆ Chapter 1: What is HTML?

Long Definition:

HTML (HyperText Markup Language) is the standard markup language used to create and structure content on the World Wide Web. It defines the structure of web pages using a system of tags and elements. HTML is not a programming language; instead, it is a markup language used to annotate text, images, and other content so that web browsers can display them properly.

It allows developers to define headings, paragraphs, images, links, forms, tables, and much more. HTML works in combination with CSS (for styling) and JavaScript (for functionality) to create complete, interactive websites.

WhatsApp Style Samjhaav:

HTML ek **designing language** hai, coding nahi. Ye batata hai ki webpage pe kya dikhana hai aur kis order me. Jaise skeleton hota hai body ka, waise hi HTML hota hai website ka.

◆ Chapter 2: Structure of an HTML Document

Long Definition:

An HTML document is structured in a tree-like format and follows a defined hierarchy. It starts with a `<!DOCTYPE html>` declaration which tells the browser the version of HTML being used (typically HTML5). The content of the page is wrapped inside `<html>` tags, which has two main sections:

1. `<head>` – Contains meta-data like title, links to CSS files, scripts, etc.
2. `<body>` – Contains the actual visible content like text, images, forms, etc.

✓ Example:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Website</title>
  </head>
  <body>
    <h1>Welcome Bro</h1>
    <p>This is my first webpage!</p>
  </body>
</html>
```

◆ Chapter 3: HTML Tags and Elements

☐ Long Definition:

Tags are predefined keywords enclosed in angle brackets (< >) that describe how the content should be displayed on the browser. An element consists of a **start tag**, **content**, and an **end tag**. For example, <p>Hello</p> is a paragraph element.

Some elements are **empty**, i.e., they don't have a closing tag, such as
 and .

◆ Chapter 4: Headings & Paragraphs

☐ Long Definition:

HTML provides six levels of headings from <h1> (largest) to <h6> (smallest). These tags are used to define titles or subtitles on a webpage. Paragraphs are created using the <p> tag, which defines a block of text.

Search engines like Google also consider heading tags important for SEO (Search Engine Optimization).

☐ WhatsApp Style:

Heading tag ka size hota hai bhai — <h1> sabse bada, <h6> sabse chhota.
Aur <p> tag ka use normal text likhne ke liye hota hai — jaise tere essay ka paragraph.

◆ Chapter 5: Formatting Tags

☐ Long Definition:

HTML includes a variety of formatting tags used to style text. These include:

- `<b>` for bold
- `<i>` for italic
- `<u>` for underline
- `<mark>` for highlighting
- `<small>` for smaller font
- `<strong>` and `<em>` for semantic importance

They improve the readability and structure of the webpage.

CODE ➡

```
<!DOCTYPE html>
<html>
<head>
  <title>Formatting Tags Example</title>
</head>
<body>

  <h1>Formatting Tags in Action</h1>

  <p><b>Yeh bold text hai</b></p>

  <p><i>Yeh italic text hai</i></p>

  <p><u>Yeh underline wala text hai</u></p>

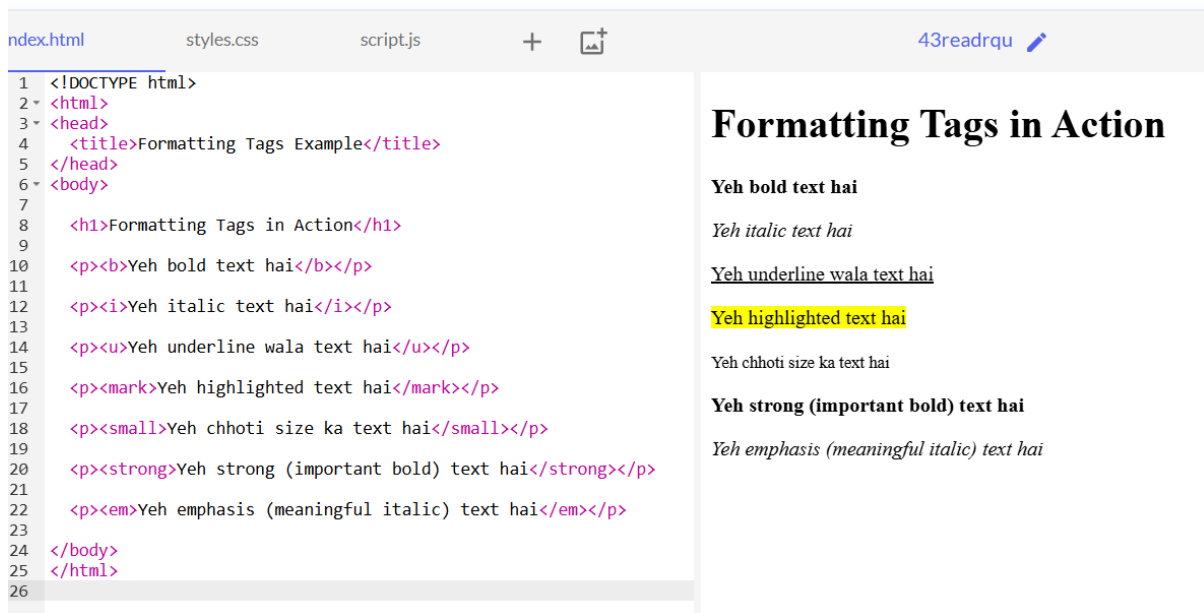
  <p><mark>Yeh highlighted text hai</mark></p>

  <p><small>Yeh chhoti size ka text hai</small></p>

  <p><strong>Yeh strong (important bold) text hai</strong></p>

  <p><em>Yeh emphasis (meaningful italic) text hai</em></p>

</body>
</html>
```



❏ WhatsApp Style:

Jaise WhatsApp me tum bold/italic karta hai message ko, waise hi HTML me bhi hota hai. Par yahan tag lagate hain — `<b>Bold</b>`, `<i>Italic</i>`

◆ Chapter 6: HTML Links and Images

📖 Long Definition:

Links are created using the `<a>` tag and are used to connect to another webpage or website. The `href` attribute specifies the destination URL. Images are embedded using the `<img>` tag, which uses the `src` attribute for the image file path and `alt` attribute for alternate text.

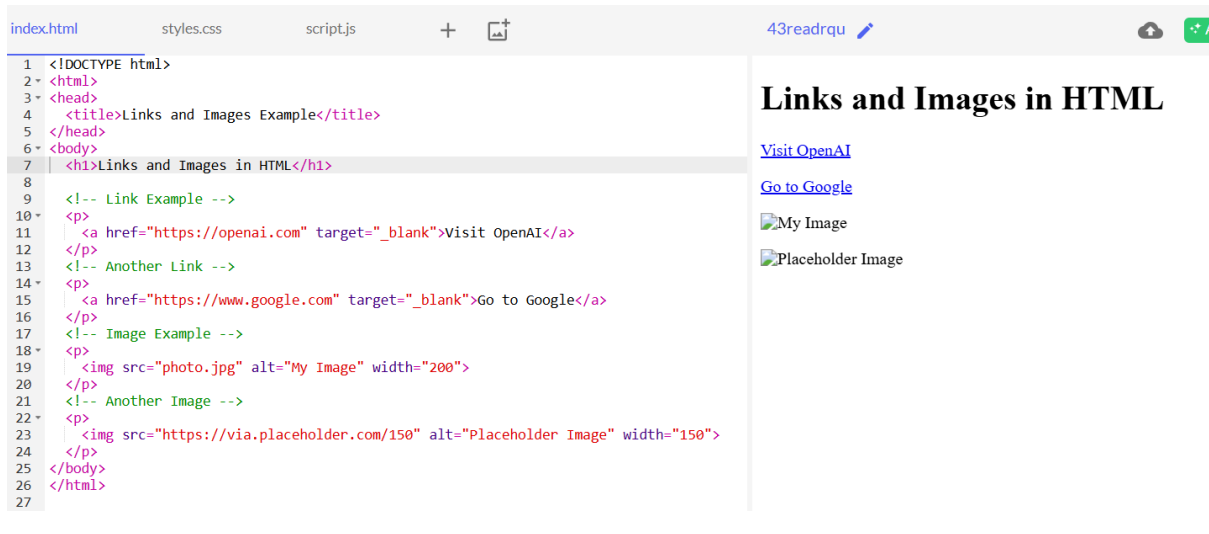
❏ WhatsApp Style:

Link = dusri site pe le jaane wala raasta
Image = dikhane ke liye, `src` = kaha rakhi hai image

✓ Example:

```
<a href="https://openai.com">Visit OpenAI</a>  

```



◆ Chapter 7: Lists (UL, OL, LI)

☐ Long Definition:

HTML supports:

- **Unordered List ()** – Items with bullets
- **Ordered List ()** – Items with numbers
- **List Item ()** – Each item in list

Used for showing grouped data like menus, tasks, or steps.

☐ WhatsApp Style:

UL = Goliyan wali list

OL = Number wali list

LI = Jo bhi point likhna hai

✓ Example:

```
<ul>
  <li>Pizza</li>
  <li>Burger</li>
</ul>
```

TASK 1 →

```
index.html  styles.css  script.js  +  43rehaxcd

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <title>Ordered and Unordered Lists</title>
6  </head>
7  <body>
8    <h1> 📄 List Examples in HTML</h1>
9
10   <h2> 📋 Ordered List (OL):</h2>
11   <p>Steps to make a sandwich:</p>
12   <ol>
13     <li>Take 2 slices of bread</li>
14     <li>Spread butter on the slices</li>
15     <li>Add vegetables or cheese</li>
16     <li>Close the sandwich</li>
17     <li>Serve and enjoy!</li>
18   </ol>
19
20   <h2> • Unordered List (UL):</h2>
21   <p>Grocery Shopping List:</p>
22   <ul>
23     <li>Milk</li>
24     <li>Eggs</li>
25     <li>Bread</li>
26     <li>Fruits</li>
27     <li>Vegetables</li>
28   </ul>
29
30 </body>
31 </html>
32
```



List Examples in HTML



Ordered List (OL):

Steps to make a sandwich:

1. Take 2 slices of bread
2. Spread butter on the slices
3. Add vegetables or cheese
4. Close the sandwich
5. Serve and enjoy!

• Unordered List (UL):

Grocery Shopping List:

- Milk
- Eggs
- Bread
- Fruits
- Vegetables

TYPE :

type value Output Style

"1"	1, 2, 3 (default)
"A"	A, B, C
"a"	a, b, c
"I"	I, II, III
"i"	i, ii, iii

```
Welcome  code_1.html  ▶  📄  📁  ...  code_1.html  ×

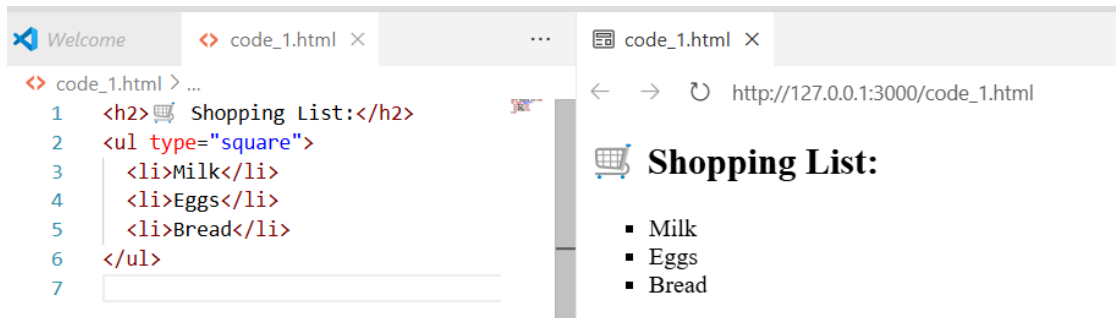
code_1.html > ...
1  <h2> 📋 To-Do List:</h2>
2  <ol type="A">
3    <li>Wake up</li>
4    <li>Brush teeth</li>
5    <li>Go to school</li>
6  </ol>
7  |end{code}

code_1.html  ×  http://127.0.0.1:3000/code_1.html

📄 To-Do List:

A. Wake up
B. Brush teeth
C. Go to school
```


type value	Bullet Style
"disc"	● Solid circle (default)
"circle"	○ Hollow circle
"square"	■ Solid square



Description List

```
<!DOCTYPE html>
<html>
<head>
  <title>Description List</title>
</head>
<body>

  <h2><img alt="HTML icon" data-bbox="178 518 198 528"/> HTML Tags - Description List</h2>

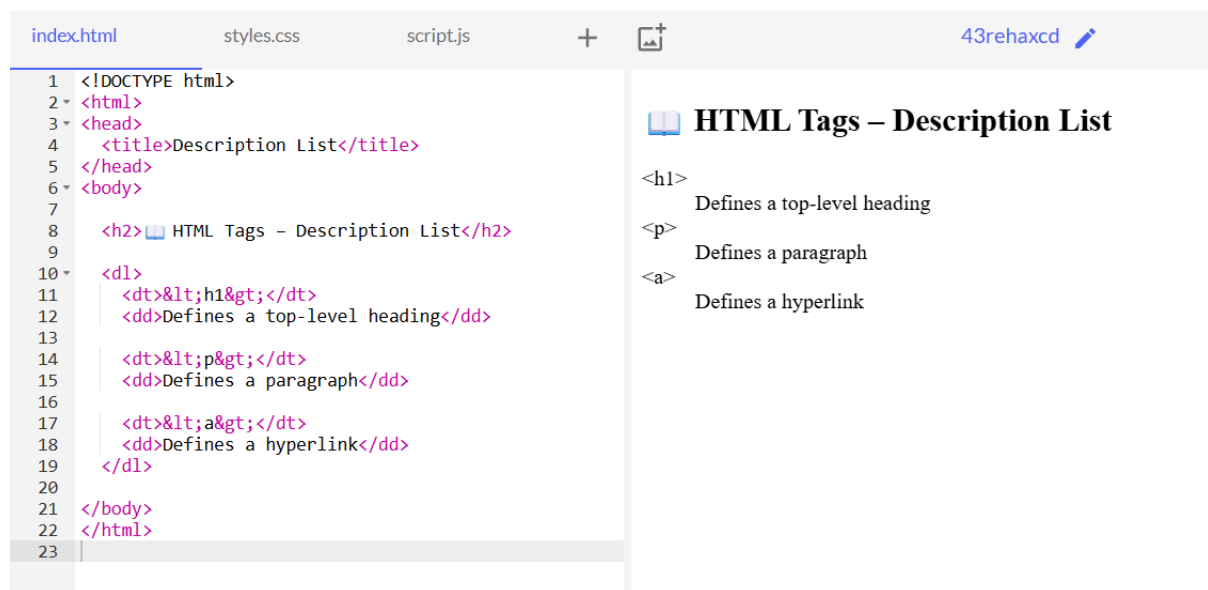
  <dl>
    <dt>&lt;h1&gt;</dt>
    <dd>Defines a top-level heading</dd>

    <dt>&lt;p&gt;</dt>
    <dd>Defines a paragraph</dd>

    <dt>&lt;a&gt;</dt>
    <dd>Defines a hyperlink</dd>
  </dl>

</body>
```

</html>



◆ Chapter 8: Tables (Updated & Complete)

📖 Long Definition (Improved):

Tables in HTML are used to organize and display data in **rows** and **columns**, similar to an Excel sheet. HTML provides specific tags for creating tables:

- **<table>**: This is the main container that defines a table.
- **<tr>** (Table Row): Defines a row in the table.
- **<th>** (Table Header): Defines the heading cell in the table. Text in **<th>** is usually bold and centered by default.
- **<td>** (Table Data): Defines the actual data cell within a row. Each **<td>** represents one cell of data under its respective column.

You can also use attributes like:

- **border**: To add borders around cells.
- **colspan** / **rowspan**: To merge multiple columns or rows.
- **cellpadding** / **cellspacing**: To control spacing inside and between cells.

Tables are useful for showing **user data, pricing, reports, attendance, etc.** in a structured way.

Rowspan ➡

index.htmlstyles.cssscript.js+🖼️

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Rowspan Example</title>
5 </head>
6 <body>
7
8   <h2>📌 Rowspan Example</h2>
9   <table border="1">
10    <tr>
11      <th>Name</th>
12      <th>Subject</th>
13    </tr>
14    <tr>
15      <td rowspan="2">Amit</td>
16      <td>Math</td>
17    </tr>
18    <tr>
19      <td>Science</td>
20    </tr>
21  </table>
22
23 </body>
24 </html>
25
```

📌 Rowspan Example

Name	Subject
Amit	Math
	Science

Colspan➡

index.htmlstyles.cssscript.js+🖼️

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Colspan Example</title>
5 </head>
6 <body>
7
8   <h2>📌 Colspan Example</h2>
9   <table border="1">
10    <tr>
11      <th colspan="2">Student Info</th>
12    </tr>
13    <tr>
14      <td>Name</td>
15      <td>Rahul</td>
16    </tr>
17  </table>
18
19 </body>
20 </html>
21
```

📌 Colspan Example

Student Info	
Name	Rahul

Cellspacing →

index.htmlstyles.cssscript.js

1 <table border="1" cellspacing="11">

2 <tr>

3 <th>Name</th>

4 <th>Age</th>

5 </tr>

6 <tr>

7 <td>Ravi</td>

8 <td>22</td>

9 </tr>

10 </table>

11

+

Name	Age
Ravi	22

CellPadding →

index.htmlstyles.cssscript.js

1 <table border="1" cellpadding="10">

2 <tr>

3 <th>Name</th>

4 <th>Age</th>

5 </tr>

6 <tr>

7 <td>Ravi</td>

8 <td>22</td>

9 </tr>

10 </table>

11

+

Name	Age
Ravi	22

□ WhatsApp Style Samjhaav:

HTML table matlab ek **digital table** – jaise aapke attendance sheet, ya mark-sheet.

- table = table start
- tr = ek puri row
- th = column ka naam (jaise "Name", "Age")
- td = asli data (jaise "Amit", "20")

th heading ke liye

td actual data ke liye

tr row ke liye

index.htmlstyles.cssscript.js+🖼️

1<table border="1">

2<caption>Student Records</caption>

3<tr>

4<th>Roll Number</th>

5<th>Name</th>

6<th>Class</th>

7<th>Marks</th>

8</tr>

9<tr>

10<td>101</td>

11<td>Anjali</td>

12<td>10th</td>

13<td>88</td>

14</tr>

15<tr>

16<td>102</td>

17<td>Rohit</td>

18<td>10th</td>

19<td>92</td>

20</tr>

21<tr>

22<td>103</td>

23<td>Sneha</td>

24<td>10th</td>

25<td>79</td>

26</tr>

27</table>

28

Student Records

Roll Number	Name	Class	Marks
101	Anjali	10th	88
102	Rohit	10th	92
103	Sneha	10th	79

✔ Example:

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Amit</td>
    <td>20</td>
  </tr>
  <tr>
    <td>Neha</td>
    <td>22</td>
  </tr>
</table>
```

❏ Output (Browser View):

Name Age

Amit 20

Neha 22

🔎 Quick Tag Summary:

Tag	Full Form	Purpose
<table>	Table	Start a table
<tr>	Table Row	Define one horizontal row
<th>	Table Header	Bold column heading

Tag	Full Form	Purpose
<td>	Table Data	Normal data cell in a table

TASK ➔

Student Information (With Rowspan & Colspan)

Roll No	Name	Class	Marks	
			Math	Science
101	Rahul Sharma	10th	88	92
102	Anjali Verma	10th	79	85

index.html
styles.css
script.js
+

43rehaxcd



```

1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Student Table with Rowspan and Colspan</title>
5  </head>
6  <body>
7    <h2> Student Information (With Rowspan & Colspan)</h2>
8    <table border="1" cellspacing="5" cellpadding="8">
9      <tr>
10       <th rowspan="2">Roll No</th>
11       <th rowspan="2">Name</th>
12       <th rowspan="2">Class</th>
13       <th colspan="2">Marks</th>
14     </tr>
15     <tr>
16       <th>Math</th>
17       <th>Science</th>
18     </tr>
19     <tr>
20       <td>101</td>
21       <td>Rahul Sharma</td>
22       <td>10th</td>
23       <td>88</td>
24       <td>92</td>
25     </tr>
26     <tr>
27       <td>102</td>
28       <td>Anjali Verma</td>
29       <td>10th</td>
30       <td>79</td>
31       <td>85</td>
32     </tr>
33   </table>
34
35 </body>
36 </html>
37

```

 Student Information (With Rowspan & Colspan)

Roll No	Name	Class	Marks	
			Math	Science
101	Rahul Sharma	10th	88	92
102	Anjali Verma	10th	79	85

◆ Chapter 9: Forms (Detailed)

Long Definition:

HTML forms collect user input and send it to a server using HTTP. Forms are made using <form> element with attributes:

- action – where to send data
- method – GET or POST

Form elements include:

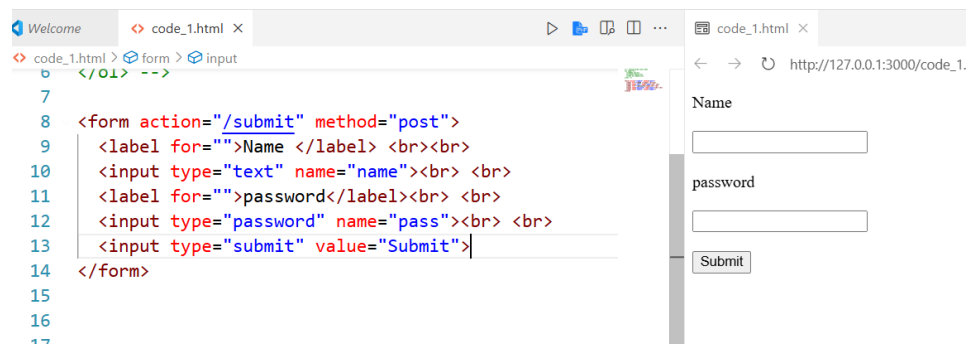
- `<input>` (text, radio, checkbox, password)
- `<textarea>` (multi-line input)
- `<select>` and `<option>` (dropdown)
- `<button>` OR `<input type="submit">`

☐ WhatsApp Style:

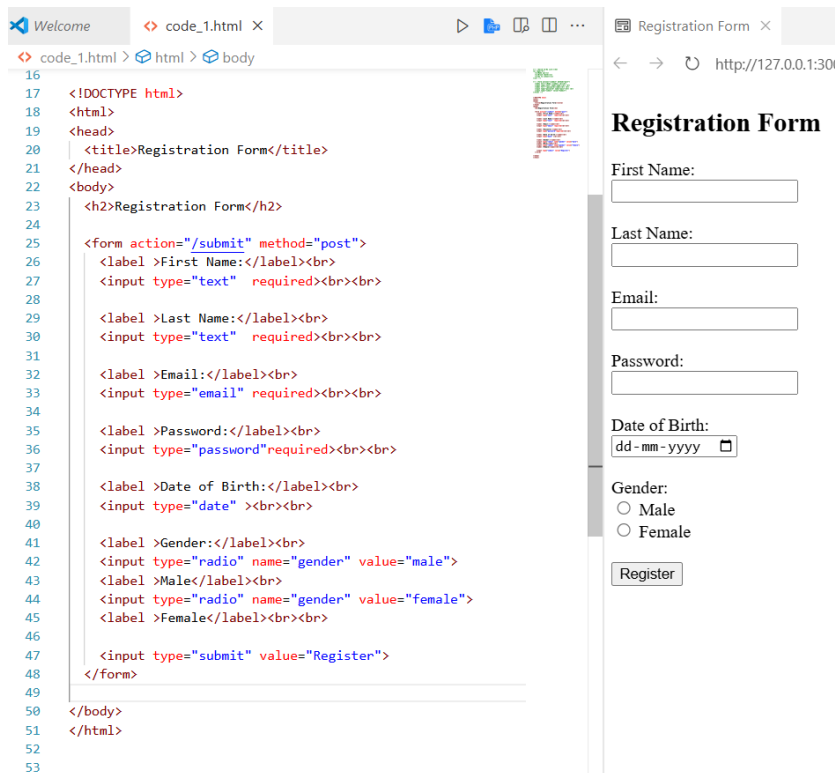
Form = wo jagah jahan Person naam, password, select kar sakta hai.
Input ka type hota hai: text, radio, password, etc.

✓ Example:

```
<form action="/submit" method="post">
  <input type="text" name="name"><br>
  <input type="password" name="pass"><br>
  <input type="submit" value="Submit">
</form>
```



Registration form (1) ➔



Registration (2) →

PRESS CLICK ME

◆ Chapter 10: Semantic Tags

📖 Long Definition:

Semantic HTML tags clearly describe their meaning both to the browser and the developer. These include:

- <header> – top part of webpage
- <footer> – bottom info
- <nav> – links navigation
- <section> – generic content section
- <article> – self-contained content like blogs

They improve accessibility, SEO, and code clarity.

📖 WhatsApp Style:

Semantic tags = samajhdaar tags 😊

Ye batate hain ki content ka kya role hai. Jaise `nav` = menu ke liye, `footer` = neeche ka info.

◆ Chapter 11: Comments in HTML

☐ Long Definition:

HTML comments are written using `<!-- comment -->` and are ignored by browsers. They help developers annotate the code for explanation, reminders, or debugging.



WEB PAGE -1

[click me](#)

WEB PAGE -2

[Click me](#)

WEBPAGE - 3

[Click me](#)

✓ THE END — Full HTML Book Notes Completed